GDAC-SA PRE-QUALIFICATION RESPONSE

Advanced Analytics Platform
Saudi Center for Geoscience Data Analytics (GDAC-SA)
Lithodat Pty Ltd
ABN: 63 627 008 904
Submitted: December 2024

هيئة المساحة الجيولوجية السعودية
Saudi Geological Survey

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EXECUTIVE SUMMARY

Lithodat Pty Ltd is honored to submit our pre-qualification response for the Geological Data Analysis Center (GDAC-SA) AI/Big Data Analytics Platform. We recognize that the Saudi Geological Survey has received approval to establish a transformational platform that will enable Saudi Arabia's Vision 2030 objective of developing the mining sector as the Kingdom's third economic pillar. This platform must integrate diverse geological data types, such as reports, geochemistry, geophysics, remote sensing, geology, and drilling records, from the National Geological Database (NGD), National Information Center (NIC), and new field campaigns, then apply advanced AI analytics to deliver mineral prospectivity predictions, target generation, and exploratory forecasting capabilities. The Saudi Geological Survey correctly identifies that much of this data exists with varying quality, incomplete metadata, inconsistent formats, and aggregations spanning decades of geological work.

We are uniquely positioned to deliver this vision because we have already built exactly what GDAC-SA envisions. Over the past decade, Lithodat together with our government, industry and academic partners, has successfully delivered two national-scale data cleaning and standardizing programs of work for federal government organizations across Australia and Canada:

- EarthBank (national geochemistry platform for AuScope Geoscience Australia)
- Isotopes.au (federal stable isotope platform developed with CSIRO and four other Australian government agencies)
- CATCH Database (Canadian Thermochronology Database for Natural Resources Canada).

These geological databases integrate the type of multi-agency, multi-decade geological data that GDAC-SA must harmonize, to power the AI-enabled analytics for mineral exploration, environmental assessment, and geoscientific research. Our track record demonstrates the capability, and proven operational delivery of the digital infrastructure GDAC-SA requires.

The Critical Foundation: Clean Data First for AI Success

The RFQ correctly identifies GDAC-SA's fundamental challenge: creating a unified geospatial analytics platform from data sources with "aggregations of data and gaps, including structured and unstructured data, some of which are without metadata." The Saudi Geological Survey recognizes that Phase 1 must create "the underlying data ecosystem" through data structure design, feature engineering, natural language processing, and new algorithm development.

This understanding is precisely correct, and we must emphasize a critical principle that determines whether GDAC-SA succeeds or fails: the only pathway to AI-enabled mineral exploration workflows is through clean, standardized, FAIR (Finable, Accessible, Interoperable and Reusable) data. Machine learning models cannot overcome poor data quality. No amount of sophisticated AI algorithms can extract reliable mineral prospectivity predictions from inconsistent terminology, missing spatial coordinates, incompatible measurement units, or uncontrolled vocabularies. Attempts to "bolt on" AI to poorly structured data inevitably fail because poor data requires more effort from AI and subsequently produces poor predictions.

Lithodat's "Clean Data First" methodology ensures that every data ingestion, transformation, and storage step creates AI-ready datasets from day one. Our proven five-stage pipeline has successfully powered three national and international data platforms, and this exact architecture will deliver GDAC-SA's requirements:

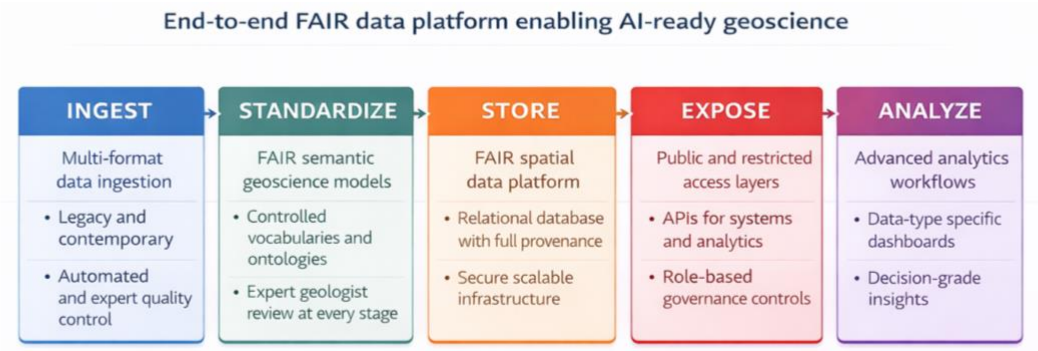


Figure 1: Lithodat's Clean First Data Pipeline

Stage 1 (Data Ingestion) employs multi-format adapters that handle Excel and CSV datasets, legacy databases, and other unstructured data with automated validation against geoscience vocabularies and quality scoring at point of entry. For the CATCH Database, we digitized 50+ years of legacy data (1970-2021) from 91 disparate sources including analogue paper reports, non-vectorized PDFs, and published figures, georeferencing sample locations from maps without GPS coordinates, exactly the historical data challenge GDAC-SA faces with NGD and NIC archives.

Stage 2 (Semantic Standardization) creates SKOS-based controlled vocabularies (ontologies) that map diverse terminology to standardized concepts, enabling AI feature engineering across heterogeneous sources. For Isotopes.au we developed 250+ standardized field mappings validated across six institutional data sources from three federal agencies, directly analogous to GDAC's challenge of harmonizing Saudi Geological Survey data across departments with different conventions. This semantic interoperability is essential for Phase 1's data ecosystem and Phase 2's AI-based mineral system modeling.

Stage 3 (FAIR-Compliant Database Storage) implements our comprehensive metadata capture, full audit trail and provenance tracking, and RESTful APIs for programmatic access by AI workflows. Our EarthBank platform achieved CoreTrustSeal certification, the international standard for trustworthy digital repositories, demonstrating our implementation of FAIR (Findable, Accessible, Interoperable, Reusable) data principles. FAIR data is not merely a compliance requirement; it is the technical foundation that enables AI success. Findability through rich metadata allows algorithms to automatically discover relevant features (mineral associations, structural controls, geochemical signatures). Accessibility via RESTful APIs enables external machine learning frameworks (PyTorch, TensorFlow, scikit-learn) to programmatically access training datasets. Interoperability through controlled vocabularies creates the semantic standardization essential for feature engineering. Reusability with comprehensive provenance tracking allows Phase 2 models to be retrained as new data arrives in Phases 3-4 without quality degradation.

Stage 4 (SGS Live Portal Development) creates dual-access systems: public-facing data portals for open geoscience access by researchers and industry, plus private secure areas for SGS internal workflows with role-based access control and user management. EarthBank and Isotopes.au are fully operational public platforms serving geoscientists globally while maintaining secure administrative features for government analytical requirements.

Stage 5 (AI/ML Application Layer) delivers advanced analytics through feature engineering from clean structured data, machine learning model training for classification /regression/clustering, mineral prospectivity mapping and target generation, and time series analysis for trend detection.

OVERVIEW

Table 1: Company Overview

RFQ Reference Number:	251140007625
Government Entity:	Saudi Geological Survey (SGS)
Applicant:	Lithodat Pty Ltd
ABN:	63 627 008 904
Submission Date:	December 2025
Submission Platform:	Etimad Portal
Prepared By:	Keith Dimech, Chief Operating Officer
Contact Email:	keith.dimech@lithodat.com
Contact Phone:	+61 428 971 396

Currency Note: *All financial values shown in Australian Dollars (AUD) and Saudi Riyals (SAR) at exchange rate: AUD 1.00 = SAR 2.45 (Monday 1st of December).*

APPLICANT INFORMATION

Application Information Form

Table 2: Advanced

Name of Applicant	Lithodat Pty Ltd
Commercial Registration Number	ABN 63 627 008 904 (Australian Business Number)
Establishment Date	25 June 2018
Years in Operation	7 years (2018-2025)
Company Type	Proprietary Limited Company (Pty Ltd)
Function	Geological Artificial Intelligence, Database and Software Services
Person Responsible	Fabian Kohlmann
Capital Value[1]	AUD $296,608 / SAR 726,690

[1] Capital Value = Net Assets as at 30 June 2025 per FY2025 Annual Report (Total Assets $430,819 - Total Liabilities $134,211 = $296,608)

Contact Information

Table 3: Address

Street Address	94 Stephensons Road
City	Mount Waverley, Victoria
Zip Code	3149
Country	Australia
P.O. Box	N/A
Telephone	+61 481 977 499
Email	fabian.kohlmann@lithodat.com
Website	www.lithodat.com

Site (Offices)	Melbourne, Australia Dublin, Ireland Hermosillo, Mexico

Company Ownership Structure

Source: Appendix A - ASIC Company Extract (02-Dec-2025)
Total Class A Shares: 1,000

Table 4: Company Ownership

#	Owner/Partner Name	Percentage	Nationality
1	Dr. Fabian Kohlmann	50.1% (501 shares)	German / Australian
2	Gerd Moritz Theile	24.0% (240 shares)	German
3	Dr. Wayne Peter Noble	14.9% (149 shares)	Australian
4	Nilesh Ambadas Vyavahare	8.0% (80 shares)	British / Indian
5	Romain Beucher	3.0% (30 shares)	French / Australian

Company Representative Information

Table 5:Company Representative

Name of Responsible Person	Fabian Kohlmann
Job Title	Chief Executive Officer (CEO)
Telephone	+61 481 977 499
Mobile	+61 481 977 499
Email	fabian.kohlmann@lithodat.com

TECHNICAL AND ADMINISTRATIVE CAPABILITIES

Previous Work Experience

Table 6: Previous Work Experience

Number of years of experience in the field of project	Lithodat - 7 years (since 2018)
Number of similar projects executed in the last three years	3 major projects Appendix G -Earthbank – Project Summary Appendix H - Isotopes.au – Project Summary Appendix I - NRCan – Project Summary

Key Performance Outcomes

Table 7: Performance Metrics

Performance Metric	EarthBank	Isotopes-AU	NRCan
On-Time Delivery	Excellent	Excellent	Excellent
Technical Quality	Excellent	Excellent	Excellent
Client Satisfaction	Excellent	Excellent	Excellent
Budget Compliance	Excellent	Excellent	Excellent

Reference Contacts

Table 8: Reference Contacts

Project	Reference Contact	Title	Email
EarthBank	Prof. Brent McInnes	AuScope Earthbank Director	b.mcinnes@curtin.edu.au
Isotopes-AU	Dr. Nina Welti	CSIRO – Environmental Isotopes Laboratory	nina.welti@csiro.au
NRCan	Dr. Jeremy Powell	Geological Survey of Canada (NRCan)	jeremy.powell@nrcan-rncan.gc.ca

Quality

Table 9: Quality Assurance

Quality Assurance Standards	Quality Management System for Geological Data Services
Evidence Provided	Quality Policy (CEO-signed)

Health Safety and Environment

Table 10: HSE Assurance

HSE Standards	Lithodat HSE Policy
Evidence Provided	HSE Policy Document (CEO-signed)

Existing Contractual Obligations

Table 11: Existing Contractual Obligations

Number of existing projects	2 active projects, 1 in Negotiation (beginning 2026)
Value of existing projects	AUD $3,800,000 combined SAR $9,310,000 combined
Status	Both projects ongoing and in good standing

Human Resources

Table 12: Human Resources

Total Number of Employees	19 (3 Management + 10 Full-time + 6 Contractors)
Number of Saudi Employees	1 (Dr. Mahdi AbuAli - GDAC Saudi Lead)
Percentage of Saudi Employees	5.3%

ADMINISTRATIVE STAFF EXPERIENCE

Table 13: Project Administration and Leadership

#	Name	Function	Specialization / Area of Expertise	Years
1	Dr. Fabian Kohlmann	Managing Director and CEO	Geoscience Leadership Strategic Planning, Business Development Former Halliburton Thermochronology Product Lead Geoscience Data Management FAIR Data Architecture	15+ years
2	Dr. Mahdi AbuAli	GDAC Saudi Lead	Global Geology Leader Saudi Government Relations, Saudi Aramco (35+ years), Executive Leadership, Petroleum Systems	35+ years
3	Keith Dimech	Chief Operating Officer	Project Management, Financial Operations, Human Relations Legal	15+ years
4	Juan Baca	Quality Manager	Data Quality Management Geology Process Leadership Database Operations Training	10+ years
5	Vinko Novak	Head of Data Security	Data Security Information Protection Compliance	20+ years

#	Name	Function	Specialization / Area of Expertise	Years
			Security Architecture	
6	Annemarie Grass	Head of Cyber Security	Cyber Security ISO 27001 Accreditation Incident Response	15+ Years

PROFESSIONAL STAFF EXPERIENCE

Table 14: Senior Technical Team

#	Name	Function	Specialization / Area of Expertise	Years
1	Dr. Wayne Peter Noble	Chief Information Officer	PhD – Thermochronology Software development: Enterprise Architecture, Cloud Systems, Agile Leadership, Software Development, Continuous Delivery Expert	25+ years
2	Dr. Qusay Abeed	GDAC Geological Technical Director	Geology & Geochemistry, Basin Modelling, CO2 Storage, Native Arabic Speaker, C++ Development, Quality Assurance Expert, Delivery Lead	18+ years
3	Gerd Moritz Theile	Chief Technology Officer	PostgreSQL Architecture, Database Design, LithoSurfer Platform, Java Development	25+ years
4	Dr. Behnam Sadeghi	ML Technical Advisor	ML/AI, Geospatial AI, Python Package Developer 2023 IAMG Vistelius Award CSIRO Research Fellow 58+ Publications	15+ years

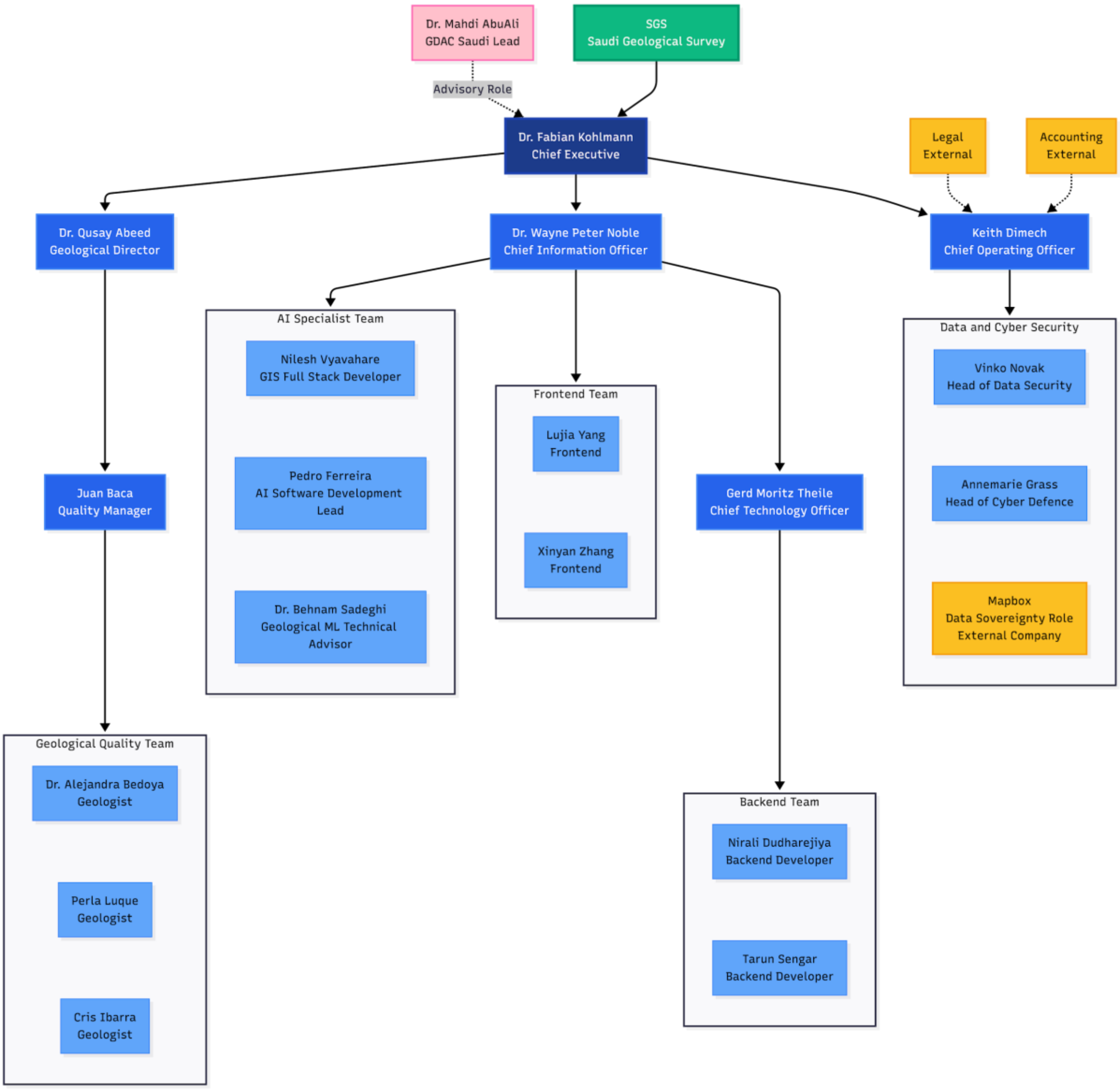
#	Name	Function	Specialization / Area of Expertise	Years
			Author on Fractals/Multifractals Fulbright Scholar	
5	Pedro Ferreira	AI Software Development Lead	Technology and AI/ML Strategy, Enterprise AI Implementation, Power Apps AI Builder	10-15 years
6	Nilesh Vyavahare	GIS Full Stack Developer	Senior Full Stack Engineer (GIS) at Principal Software Developer at Neftex - 5+ years (geological applications) Earth Observation & satellite data processing Geospatial ETL workflows (FME, Apache Airflow) ArcGIS development & plugins	18 + years

Table 15: Junior Technical Team

#	Name	Function	Specialization / Area of Expertise	Years
1	Xinyan Zhang	Frontend Developer	React, TypeScript, UI/UX Implementation, Responsive Design	3+ years
2	Lujia Yang	Frontend Developer	React, TypeScript, UI/UX Implementation, Data Science (MSc)	1+ years
3	Tarun Sengar	Backend Developer	Java, AWS, Backend Systems, API Development	5+ years
4	Nirali Dudharejiya	Backend Developer	Java, Liquibase, Backend Systems, Database Management	5+ years
5	Cris Ibarra	Geology Data Quality Specialist	Data Entry, Data Validation, Quality Control, Database Operations	3+ years

#	Name	Function	Specialization / Area of Expertise	Years
6	Perla Luque	Geology Quality Specialist	Data Entry, Data Validation, Quality Control, Database Operations	3+ years
7	Dr. Alejandra Bedoya	Geology Quality Specialist	Geochronology, Thermochronology, Tectonics, Data Validation	10+ years

PROJECT ORGANISATION CHART



EXPERIENCE IN SIMILAR PROJECTS

Project 1: EarthBank - National Geochronology Archive

Table 16: Earthbank

Project Name	EarthBank – Global Geochemistry Data Platform
Full Details	Appendix G Earthbank – Project Summary
Project Link	https://app.ausgeochem.org/ https://www.auscope.org.au
Project Location	Australia and Global
Project Content	Geological data platform integrating geochemistry and geochronology data from multiple universities, museums and research institutions across Australia. Advanced analytics for mineral exploration and tectonic history reconstruction. The platform manages over 350,000 academically verified and quality checked samples . The database serves national research infrastructure and mineral exploration initiatives internationally and across the Australian continent.
Project Owner	AuScope / NCRIS (National Collaborative Research Infrastructure Strategy)
Contract Value	AUD $2,900,000 / SAR 7,105,000
Contract Duration	5+ years (2020 - Present, ongoing)
Start Date	2020
Delivery Date	Phase 1 Delivered 2023; Ongoing enhancements
Name of Person Responsible	Dr. Fabian Kohlmann
Contact Information	Dr Brett McInnes
Email	b.mcinnes@curtin.edu.au

Project 2: Isotopes.au - National Isotopic Data Platform

Table 17: Isotopes.au

Project Name	Isotopes.au - National Isotopic Data Platform
Full Details	Appendix H - Isotopes.au – Project Summary
Project Location	Australia (National)
Project Link	https://isotopes.au/
Project Content	Platform for managing and analysing federal stable isotope data across geoscience disciplines. Integration of datasets from CSIRO, ANSTO, Geoscience Australia, and National Measurement Institute. The platform includes advanced data standardization, quality control workflows, and isotopic ratio visualization tools. Registered with AusIndustry R&D program (IISA0053835) demonstrating innovation excellence. Supporting critical minerals exploration and environmental geoscience research across Australia.
Project Owner	CSIRO / ANSTO / Geoscience Australia / NMI
Contract Value	AUD $400,000 / SAR 980,000
Contract Duration	4.5 years (July 2024 - December 2028)
Start Date	July 2024
Delivery Date	December 2028 (in progress, on schedule)
Name of Person Responsible	Dr. Wayne Noble
Client Contact Information	Dr Nina Welti - CSIRO
Email	nina.welti@csiro.au

Project 3: Canadian ThermoCHronology (CATCH) Database - Northern Canada

Table 18: NRCan CATCH Database

Project Name	Natural Resources Canada Geological Data Project
Project Location	Canada (National)
Full Details	Appendix I - NRCan – Project Summary
Project Site	https://app.lithodat.com/
Project Content	Development and delivery of Canada's national low-temperature thermochronology database for Natural Resources Canada's Geological Survey of Canada (GSC). The CATCH (Canadian ThermoCHronology) database is a comprehensive, FAIR-compliant (Findable, Accessible, Interoperable, Reuseable) platform for accessing Canadian low-temperature thermochronology (LTT) data. All Data available through Lithosurfer.
Project Owner	Natural Resources Canada (NRCan) - Canadian Federal Government
Contract Value	AUD $400,000 / SAR 980,000
Contract Duration	1.5 years (October 2021 - March 2023)
Start Date	October 2021
Delivery Date	March 2023 (DELIVERED)
Name of Person Responsible	Dr. Fabian Kohlmann
Client Contact Information	Geological Survey of Canada (NRCan) Dr. Jeremy Powell
Email	jeremy.powell@nrcan-rncan.gc.ca

FINANCIAL CAPACITY CRITERIA

6.1 Balance Sheet Statement (Last 3 Financial Years)

Table 19: Balance Sheet Statement

Balance Sheet Line Item	FY2025	FY2024	FY2023
Total Assets	AUD $430,819 SAR 1,055,506	AUD $216,906 SAR 531,420	AUD $130,477 SAR 319,669
Current Assets	AUD $364,840 SAR 893,858	AUD $176,800 SAR 433,160	AUD $93,690 SAR 229,541
Cash and Cash Equivalents	AUD $274,855 SAR 673,395	AUD $100,406 SAR 245,995	AUD $40,605 SAR 99,482
Accounts Receivable	AUD $17,487 SAR 42,843	AUD $76,394 SAR 187,165	AUD $53,001 SAR 129,852
Total Liabilities	AUD $134,211 SAR 328,817	AUD $80,012 SAR 196,029	AUD $73,574 SAR 180,256
Current Liabilities	AUD $60,404 SAR 147,990	AUD $40,106 SAR 98,260	AUD $36,787 SAR 90,128
Net Assets (Total Equity)	AUD $296,608 SAR 726,690	AUD $136,894 SAR 335,390	AUD $56,903 SAR 139,412

6.2 Financial Performance Indicators

Table 20: Financial Performance Indicators

Financial Ratio	Formula	FY2025	FY2024	FY2023
Cash Ratio	Cash / Current Liabilities	4.55	2.50	1.10
Current Ratio	Current Assets / Current Liabilities	6.04	4.41	2.55
Quick Liquidity Ratio	(Cash + Receivables) / Current Liabilities	4.84	4.41	2.54

DECLARATION

I, the undersigned, declare that:

1. All information provided in this qualification application is true, accurate, and complete to the best of my knowledge.

2. All attached documents and certificates are valid and authentic.

3. Lithodat Pty Ltd agrees to comply with all applicable laws, regulations, and requirements of the Kingdom of Saudi Arabia.

4. We understand that any false or misleading information may result in disqualification from this tender process.

5. We agree to maintain confidentiality of any information obtained during the qualification process.

6. We acknowledge that the Saudi Geological Survey reserves the right to modify dates, requirements, or cancel the RFQ process at any time.

7. If selected, we commit to registering in the Kingdom of Saudi Arabia and complying with all in-Kingdom requirements including Saudization targets.

Authorized Signatory: Dr. Fabian Kohlmann

Position: Managing Director and CEO

Company: Lithodat Pty Ltd

Date:

Signature:

LITHODAT PTY LTD

ABN 63 627 008 904

94 Stephensons Road, Mount Waverley, Victoria 3149, Australia

Email: fabian.kohlmann@lithodat.com | Website: www.lithodat.com

Phone: +61 481 977 499

Appendix A

Australian Security and Investment Commission – Lithodat Pty Ltd

Appendix B

Financial Statement FY2024-2025

Appendix C

Financial Statement FY2023-2024

Appendix D

Financial Statement FY2023-2024

Appendix E

Quality Management Policy

Appendix F

Health Safety and Environment Policy

Appendix G

EarthBank

Appendix H

Isotopes.Au

Appendix I

NRcan CATCH Database

Appendix J

Publication Summary

Appendix K

Curriculum Vitae